

Adjacent Divisibility in the Narayana Triangle

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Abstract

Let $N(n, k)$ denote the Narayana numbers. We investigate when an entry of the Narayana triangle divides its immediate right-hand neighbor. The quotient of two adjacent Narayana numbers turns out to be simultaneously a quotient of two triangular numbers. This observation leads to an exact divisibility criterion and, for every fixed column, a complete description in terms of residue classes together with an explicit natural density. The resulting row statistic, which counts nontrivial adjacent divisibilities, is the sequence [A400125](#). We show that this sequence is unbounded and obtain a lower bound for its lower asymptotic mean value. Finally, we establish a new interpretation of [A076708](#): its positive terms are precisely those rows in whose strictly increasing half two adjacent Narayana numbers occur in the ratio 2 : 1.

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1 Introduction

The Narayana numbers

$$N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}, \quad 1 \leq k \leq n,$$

form the Narayana triangle [A001263](#). They satisfy the symmetry $N(n, k) = N(n, n+1-k)$.

Divisibility properties of Narayana numbers with respect to prime numbers were studied by Bóna and Sagan [1]. Divisibility among triangular numbers has also been investigated from the perspective of partially ordered sets; see, for example, Pandey and Richman [3].

The present paper connects these two themes through a simple question: when does a Narayana number divide its immediate right-hand neighbor within the same row? The starting point is the elementary identity

$$\frac{N(n, k+1)}{N(n, k)} = \frac{(n-k)(n-k+1)}{k(k+1)} = \frac{T_{n-k}}{T_k}, \quad T_r = \frac{r(r+1)}{2}.$$

The interpretation of the quotient of adjacent Narayana numbers as a quotient of triangular numbers was suggested during an AI-assisted research process involving ChatGPT and subsequently verified by the author. Consequently,

$$N(n, k) \mid N(n, k + 1) \quad \Longleftrightarrow \quad T_k \mid T_{n-k}.$$

Thus adjacent divisibility in the Narayana triangle can be viewed as a diagonal statistic for the divisibility relation among triangular numbers. At the same time, the quotient formula admits an elementary treatment using congruences, leading to precise arithmetic descriptions.

Because of the symmetry of the Narayana triangle, it suffices to consider the strictly increasing half of each row. We therefore define

$$a(n) = \# \left\{ k : 2 \leq k < \frac{n}{2}, N(n, k) \mid N(n, k + 1) \right\}.$$

The restriction excludes the trivial divisibility at $k = 1$, where $N(n, 1) = 1$, and avoids counting the equal central pair that occurs when n is even. The sequence $a(n)$ is OEIS sequence [A400125](#) [2]. Using the triangular-number interpretation above, it may equally be written as

$$a(n) = \# \left\{ k : 2 \leq k < \frac{n}{2}, T_k \mid T_{n-k} \right\}.$$

The first values are

$$0, 0, 0, 0, 1, 0, 1, 1, 0, 1, 2, 0, 1, 3, 1, 1, 1, 1, 3, 3, 0, 1, 3, 1, 2, 3, 2, 2, 2, 1, \dots$$

2 An exact divisibility criterion

Theorem 1. *For integers $1 \leq k < n$, $N(n, k) \mid N(n, k + 1)$ if and only if*

$$k \mid n(n + 1) \quad \text{and} \quad k + 1 \mid (n + 1)(n + 2).$$

Proof. By the quotient identity, $N(n, k) \mid N(n, k + 1)$ is equivalent to $k(k + 1) \mid (n - k)(n - k + 1)$. Since $\gcd(k, k + 1) = 1$, this divisibility condition holds if and only if $k \mid (n - k)(n - k + 1)$ and $k + 1 \mid (n - k)(n - k + 1)$ simultaneously. Reducing modulo k , we obtain $(n - k)(n - k + 1) \equiv n(n + 1) \pmod{k}$. Similarly, reducing modulo $k + 1$, $(n - k)(n - k + 1) \equiv (n + 1)(n + 2) \pmod{k + 1}$. Therefore the original divisibility condition is equivalent to the two asserted conditions. $\square$

The criterion turns the original problem into a pair of elementary divisibility tests and avoids the need to compute the usually very large Narayana numbers themselves. As an illustration, consider $n = 14$. The admissible values of k are precisely $k = 2, 3, 5$. Hence $a(14) = 3$, which is the first occurrence of the value 3 in [A400125](#). Indeed,

$$N(14, 2) = 91 \mid 2366 = N(14, 3), \quad N(14, 3) = 2366 \mid 26026 = N(14, 4),$$

and $N(14, 5) = 143143 \mid 429429 = N(14, 6)$. Theorem 1 also immediately provides an efficient algorithm for computing [A400125](#).

3 Fixed columns and natural densities

For an integer m , let $\omega(m)$ denote the number of distinct prime divisors of m .

Theorem 2. *Let $k \geq 1$ be fixed. For $n \geq k + 1$, the condition $N(n, k) \mid N(n, k + 1)$ is satisfied in exactly $2^{\omega(k) + \omega(k+1)}$ residue classes modulo $k(k + 1)$. Consequently, the set of such integers n has natural density*

$$\delta_k = \frac{2^{\omega(k) + \omega(k+1)}}{k(k + 1)}.$$

Proof. Let $p^\alpha \parallel k$. Since $\gcd(n, n + 1) = 1$, the divisibility condition $p^\alpha \mid n(n + 1)$ holds if and only if $n \equiv 0 \pmod{p^\alpha}$ or $n \equiv -1 \pmod{p^\alpha}$. Thus each prime-power divisor of k contributes exactly two possibilities, and the resulting $2^{\omega(k)}$ residue classes modulo k are pairwise distinct. The same argument applies to $k + 1$: if $q^\beta \parallel (k + 1)$, then $q^\beta \mid (n + 1)(n + 2)$ holds if and only if $n \equiv -1 \pmod{q^\beta}$ or $n \equiv -2 \pmod{q^\beta}$. Therefore there are $2^{\omega(k+1)}$ pairwise distinct residue classes modulo $k + 1$. Since k and $k + 1$ are relatively prime, the Chinese Remainder Theorem combines each admissible residue class modulo k with each admissible residue class modulo $k + 1$ to produce exactly one residue class modulo $k(k + 1)$, and these combined residue classes are all distinct. The density formula follows immediately. $\square$

For example, when $k = 2$, the admissible residues are $n \equiv 1, 2, 4, 5 \pmod{6}$. Thus $\delta_2 = 2/3$. Theorem 2 completely describes the adjacent-divisibility phenomenon in any fixed column of the Narayana triangle. In particular, it shows that the problem is ultimately periodic in n , with period $k(k + 1)$.

For every $k \geq 2$, $0 < \delta_k \leq 2/3$, with equality only for $k = 2$, and $\delta_k \rightarrow 0$. Indeed, writing $f(m) = 2^{\omega(m)}/m$, we have $\delta_k = f(k)f(k + 1)$. Among two consecutive integers k and $k + 1$, one is odd and at least 3. Since $f(m) \leq 1$ for all $m \geq 2$, with equality only at $m = 2$, and $f(m) \leq 2/3$ for every odd $m \geq 3$, it follows that $\delta_k \leq 2/3$. Finally, $2^{\omega(m)} \leq d(m) \leq 2\sqrt{m}$, where $d(m)$ denotes the number-of-divisors function, so

$$\delta_k \leq \frac{4}{\sqrt{k(k + 1)}} \rightarrow 0.$$

The estimate itself is elementary, but it highlights a useful contrast: while adjacent divisibility is quite common in the first few columns, the density of such divisibilities tends to zero in any fixed column as one moves farther to the right in the Narayana triangle.

4 Average behavior of A400125

Define

$$C = \sum_{k=2}^{\infty} \delta_k = \sum_{k=2}^{\infty} \frac{2^{\omega(k) + \omega(k+1)}}{k(k + 1)}.$$

The idea of summing the column densities of Theorem 2 to obtain information about the average behavior of [A400125](#) was suggested during an AI-assisted research process involving ChatGPT and subsequently developed and verified by the author.

Proposition 3. *The series defining C converges, and*

$$\liminf_{x \rightarrow \infty} \frac{1}{x} \sum_{n \leq x} a(n) \geq C.$$

In particular, the lower asymptotic mean value of A400125 exceeds 2.

Before proving the proposition, let us note an immediate consequence: adjacent divisibilities are not a rare phenomenon. The lower asymptotic average number of nontrivial adjacent divisibilities per row exceeds two.

Proof. The classical estimate $2^{\omega(m)} \leq d(m) = m^{o(1)}$ implies $\delta_k = k^{-2+o(1)}$, and therefore the series defining C converges. Fix an integer $K \geq 2$. For each $k \leq K$, Theorem 2 shows that the condition $N(n, k) \mid N(n, k+1)$ holds with natural density δ_k . The additional restriction $k < n/2$ eliminates only finitely many values of n for any fixed k , and hence does not affect the density. It follows that

$$\liminf_{x \rightarrow \infty} \frac{1}{x} \sum_{n \leq x} a(n) \geq \sum_{k=2}^K \delta_k.$$

Since this holds for every K , letting $K \rightarrow \infty$ yields the asserted inequality. Finally, $\sum_{k=2}^{11} \delta_k = 632/315 > 2$. $\square$

A direct computation based on Theorem 1 reveals a remarkably stable picture. Let $R(x) = \max_{n \leq x} a(n)$ denote the record function of A400125.

x	mean value	$R(x)$	first occurrence
10^3	2.5930000	12	494
10^4	2.8588000	21	5984
10^5	2.9463900	27	60059
10^6	2.9751880	41	812174
10^7	2.9844491	59	6683039

Table 1: The empirical mean value of $a(n)$ and the corresponding record value $R(x)$ up to several powers of ten. The final column shows the first index at which the record is attained.

For comparison, $\sum_{k=2}^{10^7} \delta_k = 2.9887926513\dots$. The numerical data therefore suggest the natural conjecture that the inequality in Proposition 3 is in fact asymptotically sharp.

Conjecture 4.

$$\frac{1}{x} \sum_{n \leq x} a(n) \longrightarrow C.$$

A heuristic explanation is as follows. For each fixed k , Theorem 2 provides the exact density of rows contributing to $a(n)$. Proposition 3 is obtained by summing these densities over fixed columns. The numerical evidence suggests that the contribution from columns with k growing together with n becomes asymptotically negligible, so that the average value is given by the full convergent series C .

5 Arbitrarily long divisibility chains

Let $L_m = \text{lcm}(1, 2, \dots, m)$. The LCM construction used to generate arbitrarily long chains of adjacent divisibilities was suggested during an AI-assisted research process involving ChatGPT and subsequently verified by the author.

Theorem 5. *Let $m \geq 3$. If $L_m \mid (n+1)$, then*

$$N(n, 1) \mid N(n, 2) \mid \dots \mid N(n, m).$$

Proof. Let $1 \leq k < m$. Since both k and $k+1$ divide L_m , the assumption $L_m \mid (n+1)$ implies that both k and $k+1$ divide $n+1$. Consequently, $k \mid n(n+1)$ and $k+1 \mid (n+1)(n+2)$. By Theorem 1, it follows that $N(n, k) \mid N(n, k+1)$ for every $1 \leq k < m$. $\square$

For example, when $m = 6$, $L_6 = 60$. Choosing $n = 59$, we obtain $N(59, 1) \mid N(59, 2) \mid \dots \mid N(59, 6)$. Indeed, the first five consecutive quotients are 1711, 551, 266, 154, 99, corresponding to $k = 1, \dots, 5$. In particular, $a(59) = 4$.

Corollary 6. *Let $m \geq 3$. If $L_m \mid (n+1)$, then $a(n) \geq m-2$. In particular, the sequence [A400125](#) is unbounded.*

Proof. By Theorem 5, we obtain adjacent divisibilities for all

$$k = 2, 3, \dots, m-1.$$

Since L_m is divisible by both m and $m-1$, and these numbers are coprime, L_m is divisible by $m(m-1)$. Hence

$$n \geq L_m - 1 \geq m(m-1) - 1 > 2m - 2,$$

so $k < n/2$ for every $k = 2, \dots, m-1$. $\square$

Using the well-known asymptotic relation $\log L_m \sim m$, the same argument yields a lower bound for the record function $R(x) = \max_{n \leq x} a(n)$. Let m be maximal with $L_m \leq x+1$. Then $R(x) \geq a(L_m - 1) \geq m-2$. Since $\log L_{m+1} = \psi(m+1) \sim m$, it follows that

$$\liminf_{x \rightarrow \infty} \frac{R(x)}{\log x} \geq 1.$$

On the other hand, every index counted by $a(n)$ must satisfy $k \mid n(n+1)$, and therefore $a(n) \leq d(n(n+1)) = n^{o(1)}$. The true order of growth of $R(x)$ remains open, but the numerical data from Section 4 suggest substantially faster growth. In particular, the observed record values do not appear to be explained by the LCM construction alone.

6 The ratio 2 and [A076708](#)

The sequence [A076708](#) [2] consists of those integers n for which $T_n + T_{n+1} = (n+1)^2$ is itself a triangular number. The main result of this section shows that [A076708](#) admits a natural interpretation in terms of adjacent entries of the Narayana triangle. The connection between the quotient 2, the triangular-number equation $T_{n-k} = 2T_k$, and [A076708](#) was identified during an AI-assisted research process involving ChatGPT and subsequently verified by the author.

Theorem 7. *For $n \geq 1$, the following statements are equivalent:*

1. *n is a positive term of [A076708](#);*
2. *there exists an integer k with $1 \leq k < n/2$ such that $N(n, k+1) = 2N(n, k)$;*
3. *there exists an integer k such that $T_{n-k} = 2T_k$.*

Moreover, the index k is unique.

Proof. The equivalence of (2) and (3) follows immediately from the quotient identity. Next observe the identity

$$T_{n+k+1} - (n+1)^2 = 2T_k - T_{n-k}.$$

Consequently, the relation $T_{n-k} = 2T_k$ is equivalent to $T_{n+k+1} = (n+1)^2$. Since $T_n + T_{n+1} = (n+1)^2$, it follows that $(n+1)^2$ is triangular, and hence n belongs to [A076708](#).

Conversely, suppose that $n > 0$ is a term of [A076708](#). Then there exists a unique integer r satisfying $T_r = (n+1)^2$, because the triangular numbers are strictly increasing. From $T_{n+1} < (n+1)^2 < T_{2n+1}$ we obtain $n+1 < r < 2n+1$. Define $k = r - n - 1$. Then $1 \leq k < n$. Substituting into the preceding identity yields $T_{n-k} = 2T_k$. Since triangular numbers are strictly increasing, this implies $n-k > k$, and therefore $k < n/2$. Finally, the uniqueness of k follows directly from the uniqueness of r . $\square$

The first positive row indices are 5, 34, 203, 1188, 6929, 40390, $\dots$, which are precisely the positive terms of [A076708](#). For example, in row $n = 5$, $N(5, 3) = 20 = 2N(5, 2)$.

The connection with Pell equations is immediate. Writing $x = 2(n-k)+1$ and $y = 2k+1$, the condition $T_{n-k} = 2T_k$ becomes

$$x^2 - 2y^2 = -1.$$

Since this negative Pell equation has infinitely many positive integer solutions, there are infinitely many rows containing an adjacent pair of entries whose ratio equals 2. Including the initial term 0, the corresponding row indices satisfy $b_{r+2} = 6b_{r+1} - b_r + 4$, with $b_0 = 0$ and $b_1 = 5$, as recorded in [A076708](#). More generally, equations of the form $T_u = cT_v$ for a fixed multiplier c can be analyzed using Pell equations [4].

To the author's knowledge, the interpretation of [A076708](#) in terms of adjacent Narayana numbers does not appear in the existing literature. It provides a direct link between [A076708](#) and [A400125](#): the positive terms of [A076708](#) are exactly those row indices for which one of the adjacent divisibilities counted by [A400125](#) has quotient 2.

7 Concluding remarks

A striking feature of the present problem is that the simple quotient identity

$$\frac{N(n, k+1)}{N(n, k)} = \frac{T_{n-k}}{T_k}$$

leads simultaneously to exact density results, arbitrarily long divisibility chains, statistical information about [A400125](#), and a connection with Pell equations. The reformulation

$N(n, k) \mid N(n, k+1)$ if and only if $T_k \mid T_{n-k}$ shows that adjacent divisibility in the Narayana triangle may also be viewed as a natural statistic arising from the divisibility relation among triangular numbers.

Several natural questions remain open. Most notably, the numerical evidence strongly suggests that $x^{-1} \sum_{n \leq x} a(n) \rightarrow C$. A proof of this conjecture would determine the asymptotic mean value of [A400125](#) and explain the close agreement between the theoretical constant C and the numerical data of Section 4. A second open problem concerns the growth of the record function $R(x) = \max_{n \leq x} a(n)$. The preceding argument gives

$$\liminf_{x \rightarrow \infty} \frac{R(x)}{\log x} \geq 1,$$

yet the observed record values appear to grow substantially faster.

8 Declaration on the use of artificial intelligence

This project was developed within an AI-assisted research process using ChatGPT, a large language model developed by OpenAI. ChatGPT played a substantial role in the exploratory phase of the research. In particular, it contributed to the generation of mathematical ideas, the formulation of conjectures, the suggestion of proof strategies, and the identification of potential connections between Narayana numbers, triangular numbers, divisibility conditions, and Pell-type equations. Specific instances in which ideas originating from ChatGPT influenced the development of the paper are explicitly identified in the main text.

The author directed the research process, critically evaluated all suggestions generated by ChatGPT, and decided which ideas were pursued, modified, or discarded. Every mathematical statement, proof, computation, and conclusion appearing in the final manuscript was independently checked and verified by the author. The author assumes full responsibility for the correctness of all mathematical content. ChatGPT was also used to assist in the search for potentially relevant literature. All references cited in the final manuscript were independently verified by the author using the original sources or authoritative bibliographic records.

The initial working version of the research was developed in German. ChatGPT was subsequently used to identify possible issues concerning grammar, terminology, consistency, and idiomatic mathematical English. All final English wording was written and approved by the author. ChatGPT was also used to assist with manuscript preparation, LaTeX formatting, consistency checks, and compliance with journal style requirements. Such technical assistance does not affect the author's responsibility for either the exposition or the mathematical content of the paper.

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- (Concerned with sequences [A001263](#), [A076708](#), and [A400125](#).)